

2016.0 RANGE ROVER (LG), 211-00

STEERING SYSTEM – GENERAL INFORMATION

DIAGNOSIS AND TESTING

PRINCIPLES OF OPERATION

For a detailed description of the power steering system and operation, refer to the relevant description and operation section of the workshop manual.

REFER to: [Power Steering](#) (211-02 Power Steering, Description and Operation).

INSPECTION AND VERIFICATION

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If a control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- When diagnosing steering faults if issues are found to be due to interfacing system issues, e.g. anti-lock brake system, park aid, suspension, battery / charging system, etc, ensure warranty is claimed against the interfacing system where the problem is identified not steering gear.

1. Verify the customer concern

- If a road test is necessary make sure the vehicle is safe to do so

1. Using guidance from the symptom chart, and the symptoms defined in the electrical service diagnostic tool

2. Visually inspect for obvious signs of mechanical or electrical damage

Visual Inspection

MECHANICAL	ELECTRICAL
▪ Check the tires for correct pressure, size, wear and tread pattern ▪ Check for wheel rim and tire damage ▪ Check road wheel security	▪ Fuses ▪ Wiring harnesses and connectors ▪ Power steering control module

- | | |
|---|--|
| <ul style="list-style-type: none"> ▪ Check for vehicle or suspension damage ▪ Check steering ball joints and suspension components for abnormally high friction | <ul style="list-style-type: none"> ▪ Steering angle sensor module |
|---|--|

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step

4. If the cause is not visually evident, verify the symptom with the customer and refer to the approved dealer diagnostic tool.

Alternatively check symptoms against symptom chart below and check DTCs as appropriate and refer to the DTC index

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSES	ACTION
No communication with power steering control module	▪ Power steering control module power or ground circuit open circuit, high resistance ▪ High speed CAN bus (chassis) circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the power steering control module power and ground circuits for open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (chassis) circuit for short circuit to ground, short circuit to power, open circuit, high resistance
Permanently heavy/notchy steering with the engine running, even after cycling the ignition	▪ Abnormally high friction in steering ball joints or suspension components ▪ Abnormally high friction in steering or suspension components due to poor subframe mounting foot alignment or as a result of impact damage ▪ Abnormally high friction in steering gear due to water ingress	NOTE: Do not damage bellows on steering gear removal. Failure to follow this instruction may result in warranty becoming invalid for steering gear assembly. ▪ If the customer has reported notchy steering, turn the steering gear slowly across lock (with the ignition off) and

	▪ Fouling/abnormally high friction within steering column components ▪ Power steering system fault ▪ Vehicle speed signal fault	check for any tight or high friction points in the steering travel ▪ Disconnect the steering gear from the suspension. Check the suspension system for freedom of movement or abnormally high friction ▪ Check the steering and suspension system for correct installation and for signs of impact damage. For additional information on the suspension system refer to section 204-00 Diagnosis and Testing. For steering gear damage inspection see below table ▪ Remove the steering gear boots and check for evidence of water inside the steering gear boots. If evidence of water ingress is present, replace the steering gear ▪ Disconnect the steering column from the steering gear. Check the steering column and universal joints for freedom of movement or abnormally high friction. For additional information on the steering column refer to section 211-04 Removal and Installation ▪ Using the manufacturer approved diagnostic system, check the power steering control module for related DTCs and refer to the relevant DTC index ▪ Using the manufacturer approved diagnostic system, check the anti-lock brake system control module for related DTCs and refer to the relevant DTC index
Steering catch-up (heavy at fast steering wheel rates)	▪ Power steering control module power or ground circuit open circuit, high resistance ▪ Battery/charging system fault ▪ Abnormally high friction in steering ball joints or suspension components ▪ Abnormally high friction in steering or suspension components due to poor	

subframe mounting foot alignment or as a result of impact damage

- Abnormally high friction in steering gear due to water ingress
- Power steering system fault

NOTE:

The system is designed to give a heavier but stable steering feel when a low/high voltage or high temperature is detected. It is recommended to reduce harsh inputs to the steering in these scenarios, for example recently after start-up, extreme cold climate or high electrical system utilization/load. Depending on the level of reduction this could be accompanied by an alert.

- Refer to the electrical circuit diagrams and check the power steering control module power and ground circuits for open circuit, high resistance
- Refer to the relevant section of the workshop manual and test the battery and charging system
- Disconnect the steering gear from the suspension. Check the suspension system for freedom of movement or abnormally high friction. For additional information on the suspension system refer to section 204-00 Diagnosis and Testing
- Check the steering and suspension system for correct installation and for signs of impact damage. For additional information on the suspension system refer to section 204-00 Diagnosis and Testing
- Check the steering gear for freedom of movement or abnormally high friction. Remove the steering gear boots and check for evidence of water inside the steering gear boots. If evidence of water ingress is present, replace the steering gear
- Using the manufacturer approved diagnostic system, check the power steering control module for related DTCs and refer to the relevant DTC index

Permanent loss of steering assistance with engine running, even after cycling ignition	- Power steering control module power or ground circuit open circuit, high resistance - Battery/charging system fault - High speed CAN bus (chassis) circuit short circuit to ground, short circuit to power, open circuit, high resistance - Power steering system fault	- Refer to the electrical circuit diagrams and check the power steering control module power and ground circuits for open circuit, high resistance - Refer to the relevant section of the workshop manual and test the battery and charging system - Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (chassis) circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Using the manufacturer approved diagnostic system, check the power steering control module for related DTCs and refer to the relevant DTC index
Intermittent reduction/loss of steering assistance (recovers during drive cycle or after ignition cycle)	- Power steering control module power or ground circuit open circuit, high resistance - Battery/charging system fault - Power steering control module internal temperature above threshold - Power steering system fault	NOTE: The system is designed to give a heavier but stable steering feel when a low/high voltage or high temperature is detected. It is recommended to reduce harsh inputs to the steering in these scenarios, for example recently after start-up, extreme cold climate or high electrical system utilization/load. Depending on the level of reduction this could be accompanied by an alert. - Refer to the electrical circuit diagrams and check the power steering control module power and ground circuits for open circuit, high resistance - Refer to the relevant section of the workshop manual and test the battery and charging system - Check for heat shield damage or mud packing around steering gear and power steering control module. Allow

		the power steering control module to cool. Consider the environmental temperature Using the manufacturer approved diagnostic system, check the power steering control module for related DTCs and refer to the relevant DTC index
Reduction/loss of assistance when engine stop/start function activates	No fault present. This is a feature to reduce fuel consumption when the vehicle is stationary and the steering is not needed or is not recommended due to tire scrub. If steering is required, recommend prompting a re-start e.g. through clutch pedal depression	Educate customer and refer to the relevant section of the owners handbook
Excessive free play at steering wheel (when no assistance and very light torque is applied to the steering wheel)	- Steering wheel fixing loose - Steering column fixing(s) loose - Steering gear fixing(s) loose - Steering linkage toe adjust locknut(s) loose - Steering linkage outer ball joint to knuckle fixing(s) loose - Steering column pinch bolt loose - Steering column universal joint(s) excessively worn - Steering gear excessively worn	NOTE: Check for play in steering gear. Do not attempt to adjust the steering gear. Failure to follow this instruction will result in warranty becoming invalid. Refer to any relevant special service messages for checking procedures. - Check the steering wheel fixing torque - Check the steering column fixings torques - Check the steering gear fixings torques - Check the steering linkage toe adjust locknut(s) torques - Check the steering linkage outer ball joint to knuckle fixing(s) torques - Check the steering column pinch bolt torque - Check the steering column universal joints for excessive free play

		Check the steering gear for freedom of movement or abnormally high friction. Remove the steering gear boots and check for evidence of water inside the steering gear boots. If evidence of water ingress is present, replace the steering gear
Vehicle tends to pull to one side without brake applied	- Wheel/tire size incorrect - Uneven/irregular tire wear - Brake binding - Damaged or worn front/rear suspension components - Steering angle sensor fault - Abnormally high friction in steering ball joints or suspension components - Excessive wear or abnormally high friction in steering gear due to water ingress - Steering gear fixing(s) loose - Steering linkage toe adjust locknut(s) loose - Steering linkage outer ball joint to knuckle fixing(s) loose - Steering column pinch bolt loose - Lower column pinch bolt loose, steering gear retaining bolts loose - Wheel alignment incorrect	NOTE: Vehicles are optimized for pull dependant on the hand of drive and the road crown. Therefore driving on the opposite side of the road may result in pull to one side. - Check that the wheels and tires comply with the manufacturer's specification for the vehicle - Check the tires for uneven or irregular wear - Check brake system for binding - Check the front and rear suspension for damage or wear - Using the manufacturer approved diagnostic system, check the steering angle sensor module for related DTCs and refer to the relevant DTC index - Check the steering gear for freedom of movement or abnormally high friction. Remove the steering gear boots and check for evidence of water inside the steering gear boots. If evidence of water ingress is present, replace the steering gear - Check the suspension system for freedom of movement or abnormally high friction. Disconnect the steering column from the steering gear. Check the steering column and universal joints for freedom. Check the steering gear for freedom of movement - Check the steering gear fixings torques - Check the steering linkage toe adjust locknut(s) torques

		▪ Check the steering linkage outer ball joint to knuckle fixing(s) torques ▪ Check the steering column pinch bolt torque ▪ Check and adjust wheel alignment as necessary
Poor feel, steering light, not connected, requires constant correction	▪ Wheel/tire size incorrect ▪ Uneven/irregular tire wear ▪ Damaged or worn front/rear suspension components ▪ Steering angle sensor fault ▪ Abnormally high friction in steering ball joints or suspension components ▪ Excessive wear or abnormally high friction in steering gear due to water ingress ▪ Wheel alignment incorrect ▪ Power steering system fault	▪ Check that the wheels and tires comply with the manufacturer's specification for the vehicle ▪ Check the tires for uneven or irregular wear ▪ Check the front and rear suspension for damage or wear ▪ Using the manufacturer approved diagnostic system, check the steering angle sensor module for related DTCs and refer to the relevant DTC index ▪ Check the steering gear for freedom of movement or abnormally high friction. Remove the steering gear boots and check for evidence of water inside the steering gear boots. If evidence of water ingress is present, replace the steering gear ▪ Check the suspension system for freedom of movement or abnormally high friction. Disconnect the steering column from the steering gear. Check the steering column and universal joints for freedom. Check the steering gear for freedom of movement ▪ Check and adjust wheel alignment as necessary ▪ Using the manufacturer approved diagnostic system, check the power steering control module for related DTCs and refer to the relevant DTC index
Steering assistance no longer changes when terrain mode changed	▪ Terrain response system fault ▪ Body system fault ▪ Car configuration file mismatch with vehicle	

	specification	NOTE: After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the DTCs. - Using the manufacturer approved diagnostic system, check the terrain response switchpack for related DTCs and refer to the relevant DTC index - Using the manufacturer approved diagnostic system, check the central junction box for related DTCs and refer to the relevant DTC index - Using the manufacturer approved diagnostic system, check and update the car configuration file as necessary. Clear the DTCs and retest
Park assist function is not working	- Tire pressure(s) incorrect - Wheel/tire size incorrect - Damaged or worn front/rear suspension components - Wheel alignment incorrect - Fouling/abnormally high friction within steering wheel/steering column components - Parking aid system fault - Body system fault - Power steering system fault - Steering angle sensor fault - Anti-lock brake system fault - Transmission system fault - Car configuration file mismatch with vehicle	NOTE: After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the DTCs. - Check and adjust the tire pressures as necessary - Check that the wheels and tires comply with the manufacturer's specification for the vehicle - Check the front and rear suspension for damage or wear - Check and adjust wheel alignment as necessary - Disconnect the steering column from the steering gear. Check the steering wheel/steering column and universal joints for freedom of movement or abnormally high friction - Using the manufacturer approved diagnostic system, check the parking

	specification	aid control module for related DTCs and refer to the relevant DTC index ■ Using the manufacturer approved diagnostic system, check the central junction box for related DTCs and refer to the relevant DTC index ■ Using the manufacturer approved diagnostic system, check the power steering control module for related DTCs and refer to the relevant DTC index ■ Using the manufacturer approved diagnostic system, check the steering angle sensor module for related DTCs and refer to the relevant DTC index ■ Using the manufacturer approved diagnostic system, check the anti-lock brake system control module for related DTCs and refer to the relevant DTC index ■ Using the manufacturer approved diagnostic system, check the transmission control module for related DTCs and refer to the relevant DTC index ■ Using the manufacturer approved diagnostic system, check and update the car configuration file as necessary. Clear the DTCs and retest
Shimmy through steering wheel without brake applied	■ Tire damage ■ Uneven/irregular tire wear ■ Wheel imbalance ■ Drive shaft out of balance (normally accompanied by more severe body shake and less steering wheel feedback compared to a shimmy) ■ Damaged or worn front/rear suspension components	NOTE: Shimmy: Defined as a lower frequency feedback/oscillation through the steering, which changes with vehicle speed. A good example of this is when the wheels are out of balance at around 65-85 kph ■ Check the tires for damage ■ Check the tires for flatspots and uneven or irregular wear ■ Check the wheels for imbalance ■ Check the drive shafts

		Check the front and rear suspension for damage or wear
Harsh, high frequency vibration through steering wheel	- Tire pressure(s) incorrect - Wheel/tire size incorrect - Power steering system fault	- Check and adjust the tire pressures as necessary - Check that the wheels and tires comply with the manufacturer's specification for the vehicle - Using the manufacturer approved diagnostic system, check the power steering control module for related DTCs and refer to the relevant DTC index
Vehicle turns hard into end lock / lock stops	- Steering angle sensor fault - New steering components installed and calibration routine not performed - Uneven toe setting (tyre contact with suspension or body components)	- Using the manufacturer approved diagnostic system, check the steering angle sensor module for related DTCs and refer to the relevant DTC index - Using the manufacturer approved diagnostic system, perform routine - Power Steering Soft Lock End Stops Reset Routine - Check and adjust wheel alignment as necessary
Knocking through steering wheel when pulling away and/or braking with no warning light	- Steering gear fixing(s) loose - Steering column pinch bolt loose - Excessive wear in steering system - Steering wheel/steering column components fixings loose - Damaged or worn front/rear suspension components	NOTE: In the case steering gear bolts are loose, often a creaking is heard when steering near the lock stops position. - Check the steering gear fixings torques - Check the steering column pinch bolt torque - Check steering column components for excessive wear and rectify as necessary - Check the steering wheel/steering column fixings torques - Check the front and rear suspension for damage or wear

Scraping from inside of vehicle when turning the steering wheel	- Steering column fouling components - Clockspring mechanism fouling - Uneven toe setting (tyre contact with suspension or body components)	- Check the steering column for fouling on cowl, carpets, or harnesses - Check the clockspring mechanism for security - Check and adjust wheel alignment as necessary
Excessive noise (refer to further applicable special service message information) with no warning light, instability, graunch	- Tire pressure(s) incorrect - Wheel/tire size incorrect - Damaged or worn front/rear suspension components - Foul condition - Water ingress to the steering gear - Clockspring internal failure	- Check and adjust the tire pressures as necessary - Check that the wheels and tires comply with the manufacturer's specification for the vehicle - Check the front and rear suspension for damage or wear - Check for a foul condition between the steering column and the body. Check for a foul condition between the steering gear and the body - Check the steering gear for freedom of movement or abnormally high friction. Remove the steering gear boots and check for evidence of water inside the steering gear boots. If evidence of water ingress is present, replace the steering gear - Install a new clockspring
Excessive noise (refer to further applicable special service message information) with amber warning light, instability, graunch	- Tire pressure(s) incorrect - Wheel/tire size incorrect - Damaged or worn front/rear suspension components - Foul condition - Water ingress to the steering gear - Clockspring internal failure - Transmission system fault - Power steering control module is not configured correctly	- Check and adjust the tire pressures as necessary - Check that the wheels and tires comply with the manufacturer's specification for the vehicle - Check the front and rear suspension for damage or wear - Check for a foul condition between the steering column and the body. Check for a foul condition between the steering gear and the body - Check the steering gear for freedom of movement or abnormally high friction. Remove the steering gear boots and check for evidence of water inside the steering gear boots. If evidence of

	▪ Abnormally high friction in steering or suspension components due to poor subframe mounting foot alignment or as a result of impact damage	water ingress is present, replace the steering gear ▪ Install a new clockspring ▪ Using the manufacturer approved diagnostic system, check the transmission control module for related DTCs and refer to the relevant DTC index ▪ Using the manufacturer approved diagnostic system, re-configure the power steering control module with the latest level software ▪ Check the steering and suspension system for correct installation and for signs of impact damage
Excessive noise (refer to further applicable special service message information), zipper, whoop, rattle	▪ Foul condition ▪ Power steering control module power or ground circuit open circuit, high resistance ▪ Battery/charging system fault ▪ Steering gear wear excessive ▪ Abnormally high friction in steering or suspension components due to poor subframe mounting foot alignment or as a result of impact damage ▪ Uneven toe setting (tyre contact with suspension or body components)	▪ Check for a foul condition between the steering column and the body. Check for a foul condition between the steering gear and the body ▪ Refer to the electrical circuit diagrams and check the power steering control module power and ground circuits for open circuit, high resistance ▪ Refer to the relevant section of the workshop manual and test the battery and charging system ▪ Install a new steering gear ▪ Check the steering and suspension system for correct installation and for signs of impact damage ▪ Check and adjust wheel alignment as necessary

ELECTRIC POWER STEERING GEAR INSPECTION - WITH HEAVY/NOTCHY STEERING GEAR

NOTE:

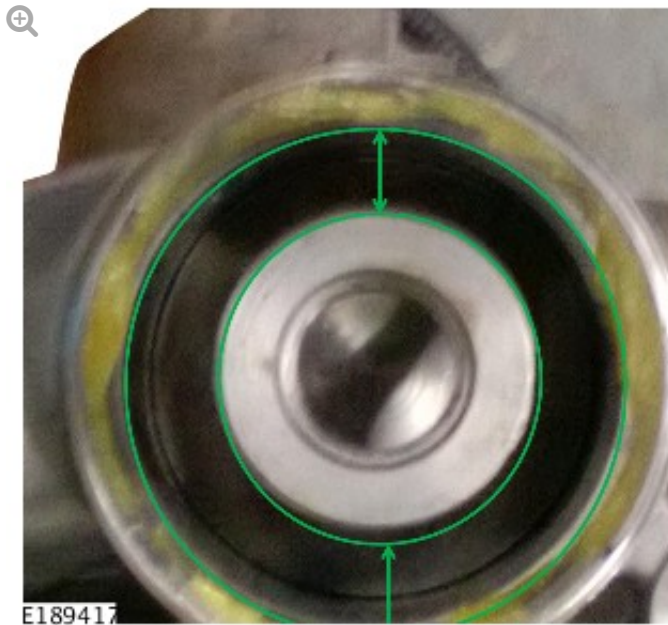
Some variation in the illustrations may occur, but the essential information is always correct.

Electric Power Steering Gear Damage Assessment Checklist

- 1.** Log VIN
- 2.** Confirm noise, is it repeatable on the vehicle?
- 3.** Has vehicle previously had accident damage repairs?
- 4.** Are there any signs of impact damage to the wheels/tires?
- 5.** Is the steering free to turn lock to lock on a wheel free ramp?
- 6.** Is front steering alignment correct?
- 7.** Does the noise occur on all surfaces?
- 8.** Does the noise occur in all temperatures?

ELECTRIC POWER STEERING GEAR ACCIDENT DAMAGE CHECK

The example below shows a good steering gear in central position.



Bent Steering Gear

Below are some examples of damage that would not be accepted under the terms of the Jaguar Land Rover warranty agreement. Please note: these are examples only and do not represent all warrantable/non warrantable customer concerns.

Check the differences between the distances at point A & B. Check if the rack is central and not rubbing on the end stop damper.

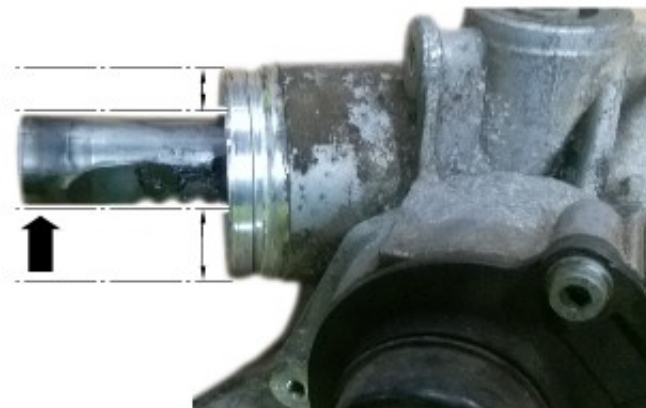


The rack clearly shows that it is not centralized meaning it is a bent steering

gear issue.



E189419

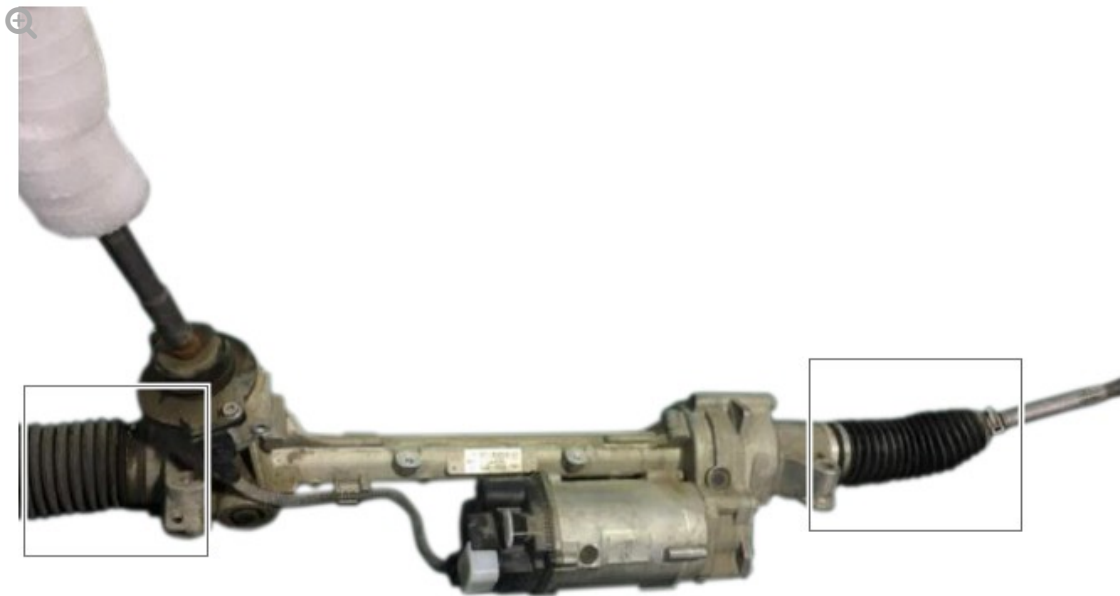


E189420

The rack is not centralized to the gear housing meaning it is a bent steering gear issue.

Sign of Recent Repair

Check if the tie rod ends were recently changed. Visually inspect if the conditions of the boot are the same (dirty or clean).



E189421

See the difference between the boots highlighted in yellow and in red. This will tell you if a tie rod has been replaced before in which case you can suspect accident damage.

Water/Debris Damage

Check if boot clips are in position and they were not recently disturbed or moved. Check if clips are the correct type.



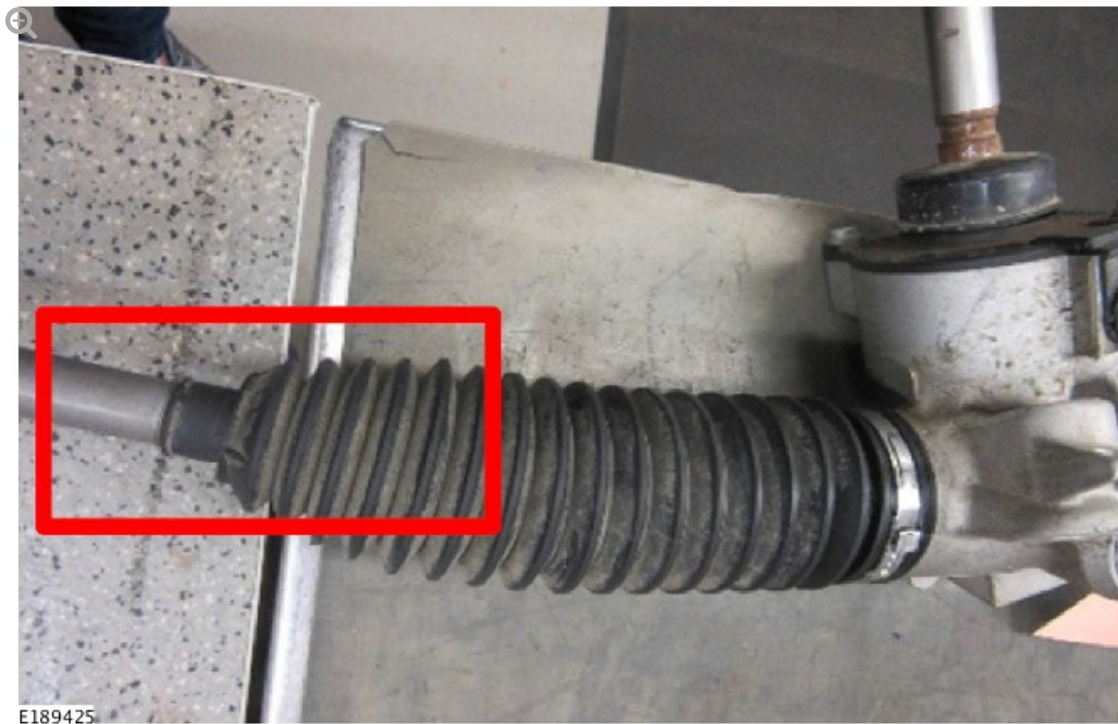
E189422



E189423



E189424



E189425

Check for damages and pin holes on the boots.



E189426



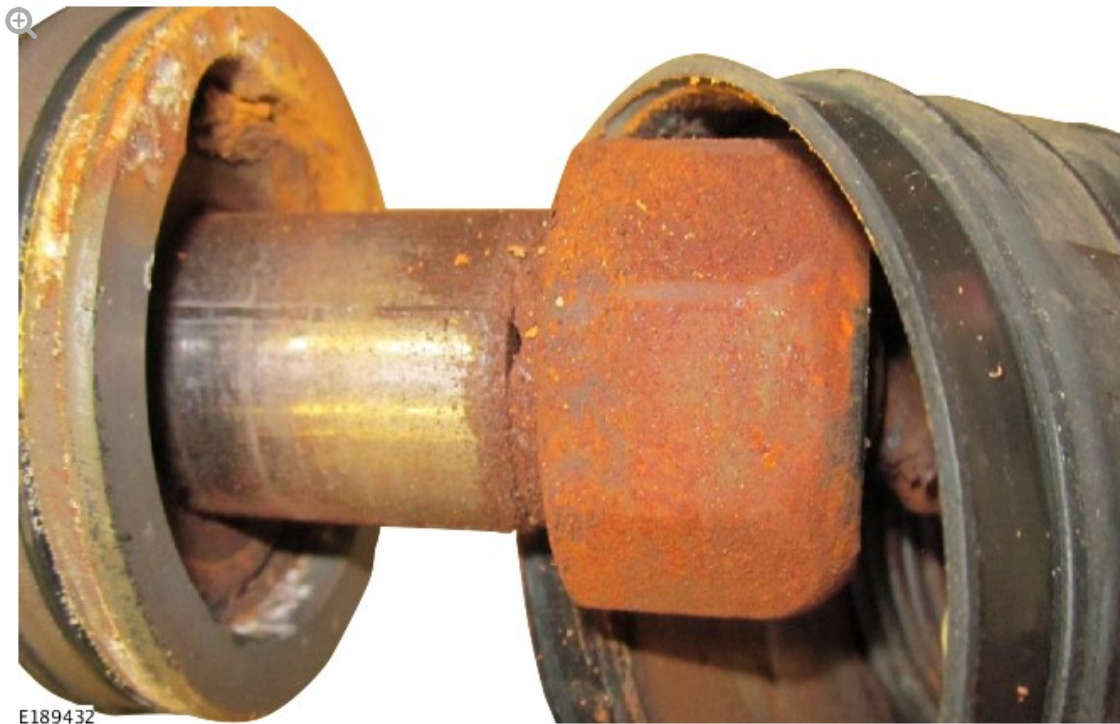
Check for split boots. Check for repair marks on tie rods.





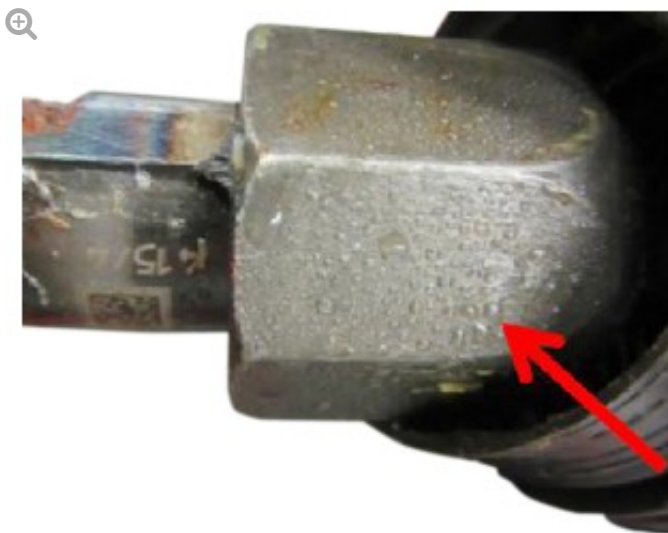
Is there evidence of water/debris into steering gear.





E189432

Check for Rust, condensation, and water deposits.



E189433



ECU Damages

Check for cracks on PSCM housing.



E189436



E189437

Check for scratches and damages on PSCM connectors and the steering gear.



E189438



E189439

Damage to Connector, Seals or Wiring

Is there evidence of damage to connector clips.

Check for removed/damaged connectors.



Check for Harness damage between ECU and Torque sensor.



Impact Damage to Motor/Gear Housing

Check for evidence of impact on the motor/gear housing.



Check the alignment of the bolt between the housing and the motor and external damages on the motor housing that can alter belt tension.





E189446

Drain Plug (where fitted)

Is Water Drain Valve deployed?



E189447

If Water Drain Valve is deployed, that means water is inside the gear, check for damaged bellows and clips.

Damage of Seat Material

Is there evidence of damage to seat material?



Damage to Tie Rod

Is there evidence of tie rod bending?



If there is heavy steering and bent tie rod there is a clear indication that the vehicle was subjected to heavy impacts.

Damage to Ball Joint

Is there evidence of damage to ball joints?



Summary

WARNING:

The following issues are not covered under Warranty. Dealers should not claim these issues.

- **Accident damage**
- Bent Steering Gear
- Bent Tie Rod
- Damage PSCM housing & Power Motor
- **Water damage**
- Rusty gears
- Water ingress

DTC INDEX

For a list of Diagnostic Trouble Codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00.

REFER to: [Diagnostic Trouble Code Index - DTC: Power Steering Control Module](#) (100-00 General Information, Description and Operation).